A J Rivas hux 960

Pre-Lab Questions

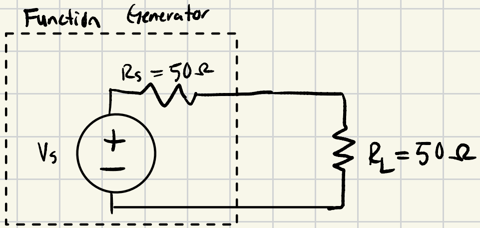
1. Given that the input impedance of the function generator is 50Ω, and using a load resistance of 50Ω, what is the voltage across the load if the function generator's internal voltage source provides $10V$? In this situation you are following the 50Ω load assumption. What voltage would you expect to see on the display on the function generator (no wrong answer)? Hint: Set the problem up as a voltage divider circuit.

2. A 1x probe will provide a $1M\Omega$ impedance. If we were to measure the signal coming from the function generator directly using the oscilloscope probes, we would not be following the 50Ω load assumption. To calculate the measured voltage we would receive we would need to set the problem up as a voltage divider circuit with a $1M\Omega$ ohm load resistance. What is the voltage across the probe terminals (i.e., across the $1M\Omega$ ohm load) if the function generator is providing $10V$? In this situation, what voltage would you expect the display on the function generator to display (no wrong answer)?

Note that for both 1 and 2 the calculated voltage is different across the load and the probe, although the function generator is set to the same $10V$ value, and hence the display would also be the same (whatever value it may show) in both situations. Take this into consideration when you procure and evaluate the meaning between the last measurements taken in Parts A and B.

[Provide the Pre-Lab answers separate from the lab report, and show your work.]

1)



$$V_L = \frac{R_L}{R_L + R_s} \cdot V_s \qquad V_L = \frac{50\Omega}{50\Omega + 50\Omega} \cdot 10V \qquad \underline{V_L = 5\ V}$$

The Voltage seen on the display would be $\underline{10V}$.

2) $$V_L = \frac{1M\Omega}{1M\Omega + 50\Omega} \cdot 10\ V$$

$$V_L \approx 9.99V \approx \underline{10V}$$

The Screen would shown around $\underline{10\ V}$.